Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Neonatal Infection
Eastern and Northern Perinatal Services

SUMMARY

This guideline outlines the process for a prompt diagnosis, monitoring and treatment of neonatal infection.

KEY POINTS

The essential elements of this guideline are:

- Recognising and diagnosing early onset neonatal infection
- Monitoring, investigating and treating early onset neonatal sepsis
- Neonatal Oral Antibiotics at Home (NOAH)
- Recognising and diagnosing Late onset neonatal sepsis
- Monitoring, investigating and treating late onset neonatal sepsis
- Meningitis
- Monitoring, investigating and treating Meningitis
- Neonatal Antibiotic Guidelines

CONTENTS

Section		Page
1.	**Introduction**	3
2.	**Background**	3
3.	**Definitions**	3
4	**Presenting Features**	4
5	**Early Onset Neonatal Sepsis**	4
5.1	Neonatal Oral Antibiotics at Home (NOAH)	8
6	**Late-Onset Neonatal Sepsis**	11
7	**Meningitis**	13
8	**Neonatal Antibiotic Guidelines**	16
5	**Monitoring Compliance with this Guideline**	17
6	**References**	17
7	**Publication Details**	18
Appendices		
Appendix 1: Kaiser Neonatal Early-Onset Sepsis Calculation		19
Appendix 2: Neonatal Oral Switch Telephone Review Proforma		20
Appendix 3: Referral to Audiology		22

1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 Sepsis is common in the neonatal period, and prompt recognition, escalation and treatment reduces morbidity and mortality, and reduces separation of mother and baby.

2. BACKGROUND

2.1 Immunological naivety, exposure factors and immaturity of host defences render neonates very susceptible to infection. The risk of sepsis in neonatal unit admissions is magnified by factors linked to prematurity and neonatal intensive care. Neonatal sepsis often develops rapidly with high mortality risk in the absence of specific therapy, demanding early empirical antibiotic therapy when sepsis is suspected.

Aseptic handling of central venous catheters and adherence to hand-washing before and after patient contact can effectively reduce nosocomial infection in preterm infants.

3. DEFINITIONS

3.1 The term sepsis incorporates ‘bacteraemia, pneumonia, meningitis, urinary tract, bone and joint infection. Common presenting features do not reliably localise the site of infection, and any of these may co-exist.

Early onset sepsis – present before 72hrs of life.

Late onset sepsis – present after 72hrs of life.

4. PRESENTING FEATURES

(Beware: all individual features lack sensitivity or specificity).

Red flags: (NICE)

- Apnoea
- Seizures
- Need for cardiopulmonary resuscitation
- Need for mechanical ventilation
- Signs of Shock

- Respiratory
  - Respiratory distress (also see grunting baby guideline)
  - Apnoea
  - Hypoxia – oxygen saturation <90% if <36weeks, or <95% if ≥36 weeks gestation
  - Term infant requiring ventilation
- Cardiovascular
  - Signs of shock
  - Bradycardia (<100/min) or Tachycardia (>160/min)
  - Persistent pulmonary hypertension (PPHN)
- Haematological
  - Unexplained excessive bleeding, thrombocytopenia or abnormal clotting
  - Neutropaenia
- Neurological
  - Seizure
  - Signs of encephalopathy
  - Altered behaviour or responsiveness
  - Lethargy
  - Hypotonia
- Abdominal
  - Feeding difficulties
  - Feed intolerance, including vomiting, excessive gastric aspirates and abdominal distension
- Other
  - Temperature instability (<36ºC or >38ºC)
  - Jaundice (especially in the first 24hrs)
  - Rash
  - Hyper/hypoglycaemia
  - Metabolic acidosis (BE less than -10 mmol/L) – infant blood gas, not cord gas
  - Unexplained resuscitation requirement at birth
  - Non-specific parental or nursing concern

5. EARLY-ONSET NEONATAL SEPSIS

Organisms:

Early onset sepsis - Organisms responsible for early onset sepsis come from the maternal genital tract. The predominant pathogens are group B streptococcus (GBS), E coli, other Streptococci, Haemophilus influenzae, other anaerobes and Listeria monocytogenes.

Antenatal prevention:

At-risk women should be offered intrapartum IV antibiotic prophylaxis (See 'Group B Strep – Prevention of Early Onset GBS')

Risk factors for early onset neonatal infection:

Red flag risk factor

🚩 Suspected or confirmed infection in another baby in the case of multiple pregnancy

Risk factors

- Invasive group B streptococcal infection in a previous baby or maternal group B streptococcal colonisation, bacteriuria or infection in the current pregnancy. (See GBS guideline here)
- Pre-term birth following spontaneous labour before 37 weeks' gestation.
- Confirmed rupture of membranes for more than 18 hours before a pre-term birth.
- Confirmed prelabour rupture of membranes at term for more than 24 hours before the onset of labour.
- Intrapartum fever higher than 38°C if there is suspected or confirmed bacterial infection.
- Clinical diagnosis of chorioamnionitis.

5.1 Differential Diagnosis:

Metabolic conditions, congenital heart disease and perinatal viraemia (especially herpes simplex and entroviruses) can present with clinical features indistinguishable from neonatal sepsis.

4.1.3 When to Monitor, Investigate and Treat Infants

Clinical concern always takes priority – any baby clinically suspected of having sepsis must be screened and treated irrespective of NICE or the EOS calculator (Kaiser).

It is responsibility of the midwives to identify risk factors or presenting features and communicate with the neonatal team.

Neonatal staff should assess the baby as soon as possible (within 1 hour), if this is not possible then a verbal plan should be made within the hour.

Babies ≥34 weeks gestation with any risk factors require an Early Onset Risk Assessment using the Kaiser calculator https://neonatalsepsiscalculator.kaiserpermanente.org/

Babies <34 weeks with two risk factors or one red flag should be started on antibiotics. (Do not use Kaiser)

Flow Sheet for Suspected Early Onset Sepsis (babies <72 hours old)

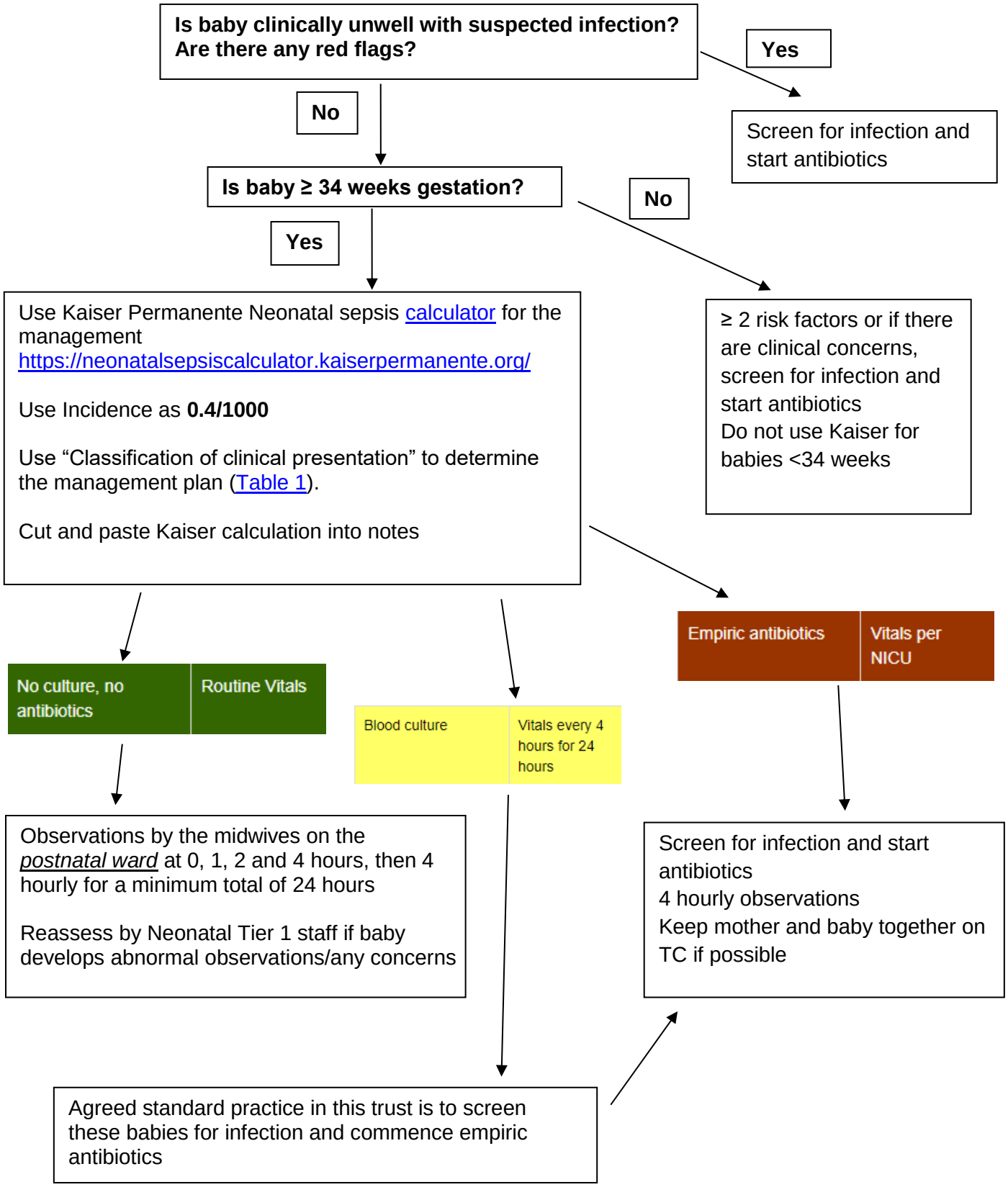


TABLE 1.0 – Kaiser Permenante Definitions of Well, Equivocal and Clinically Unwell

Appears Well	No persistent physiological abnormalities
Equivocal	Tachycardia (>160), tachypnoea (>60), temperature instability (<36.5 >38), Respiratory distress (Grunting*, recession, Flaring) but not requiring Oxygen One physiological abnormality persisting for >4 hours OR Two or more physiological abnormalities lasting for > 2 hours *– see Grunting Baby guideline
Clinical Illness	Persistent need for High Flow, CPAP or Ventilation Haemodynamic Instability Neonatal Encephalopathy including seizures, APGAR <5 at 5min Need for supplementary Oxygen to maintain oxygen saturations

4.1.4 Investigations:

Standard Investigations:

- Blood culture – minimum recommended volume 0.5ml. Must be taken prior to starting antibiotics
- Full blood count
- CRP (C-Reactive Protein) – repeated at 18-24hr post presentation.

Supplementary investigations:

- Blood gas and Chest X-Ray – required in all infants with respiratory distress or a continuing oxygen requirement
- Lumbar puncture – see Neonatal Meningitis section
- Urine culture – not routinely required in suspected early onset sepsis
- Eye swab – consider bacterial swab and COBAS swab (for chlamydia) in those with purulent eye discharge (see sticky eye guideline)
- Umbilical swab – bacterial swab only if periumbilical cellulitis present

If the baby is clinically unwell, other investigations need to be considered depending on the symptoms present e.g. calcium, clotting studies, LTFs, bilirubin, U&Es.

4.1.5 Treatment

Initial Management:

General

Close support and monitoring are essential. All neonates commencing antibiotics require close observation (formal observations every 4 hours), and a Tier 2 medic or equivalent should be informed.

If feasible and baby is clinically well, consider inserting cannula and administering antibiotics without separating mother and baby (e.g. Insert cannula on resuscitaire on labour ward). A cannulation trolley for this is stored in the NNU library. A neonatal nurse may be available to assist and administer antibiotics.

Antibiotics

The first dose of antibiotics should be given within 1 hour (ideally within 30 minutes) of the decision to treat. Do not wait for results first.

Empirical first line antibiotics for suspected early onset infection are

- IV Benzylpenicillin 25mg/kg 12 hourly
- IV Gentamicin 5mg/kg 36 hourly

See 'Neonatal Antibiotics' section for more details

Ongoing Management:

Stopping antibiotics

Wherever possible, antibiotics should be stopped at 36 hours if

- Baby is clinically well
- CRPs are low
- Cultures negative

Persisting Symptoms or Signs - Consider HSV Infection

Symptoms may include:

- Fever, requiring paracetamol
- Abnormal liver or renal function
- Unusual rashes

Consider adding Acyclovir even if there is no history of Herpes and no positive culture or PCR.

See Clinical Guideline for Genital Herpes in Pregnancy and BASH/RCOG national guideline

4.2 Neonatal Oral Antibiotics at Home (NOAH)

4.2.1 Eligibility for NOAH

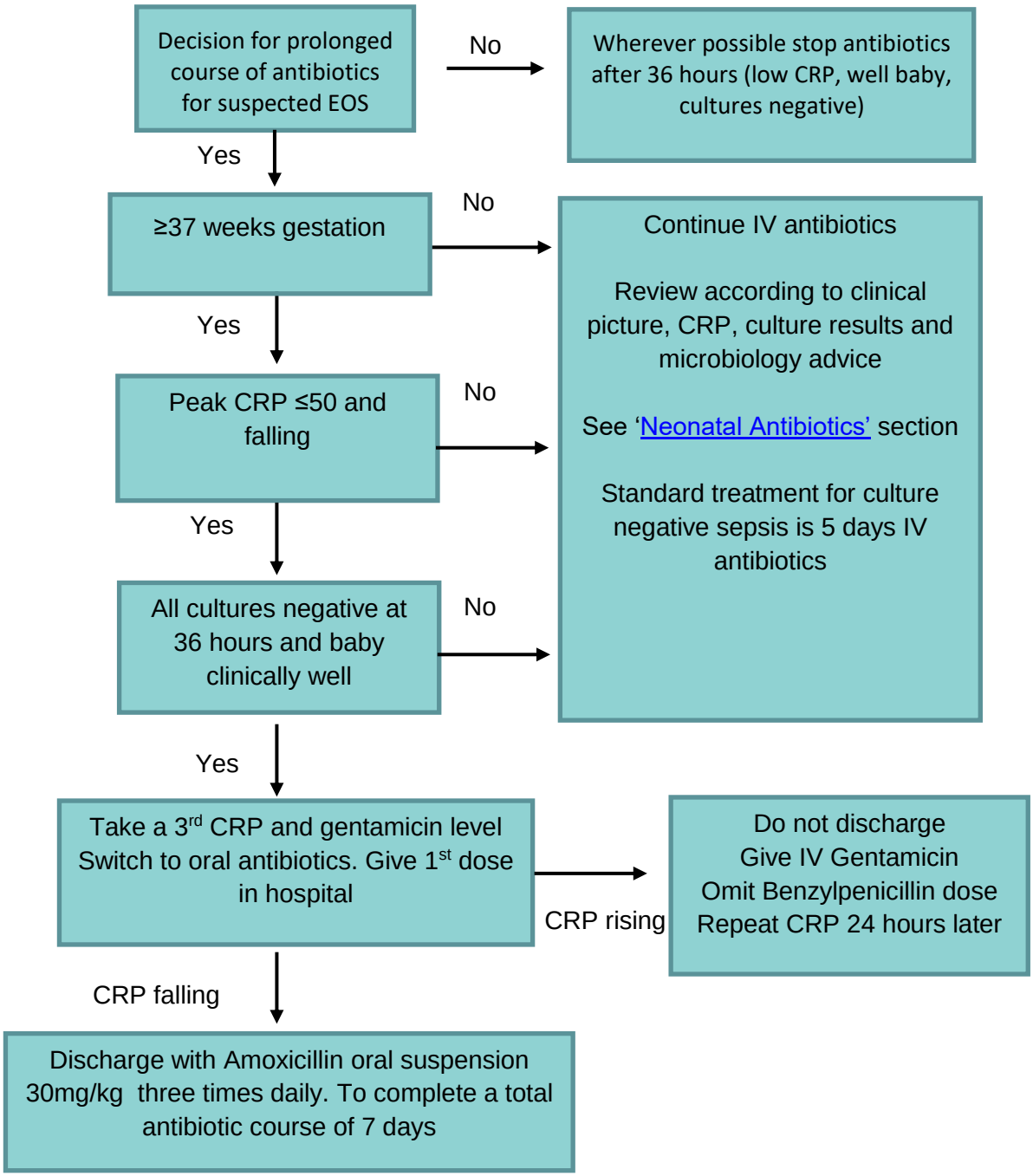
For babies >37 weeks gestation with **suspected early onset infection** for whom there are ongoing concerns about infection (e.g. elevated CRP, persistent clinical signs) a prolonged course of antibiotics may be indicated.

For babies with early onset sepsis who meet the criteria (see oral antibiotic flowsheet), a switch to oral antibiotics and early discharge from hospital may be considered.

IV antibiotics should be continued in babies who are clinically unwell, have a positive blood culture, have a peak CRP of >50 or a CRP that is not falling or are unable to tolerate oral medication.

Additional information about NOAH is available at NOAH (Neonatal Oral Antibiotics at Home) - Health Innovation South West

4.2.2 Oral Antibiotic Flowsheet



4.2.3 Ongoing Management

If peak CRP result is than 50 and falling ,and all cultures are negative at 36 hours, babies are required to complete a total antibiotic course of 7 days (this includes the switching of antibiotics from IV administration to oral). Providing they are clinically well, their IV antibiotic treatment can be switched to oral antibiotics at 36 hours.

The first dose of oral antibiotics must be given prior to discharge of the baby to ensure that this is tolerated. They should be observed for a minimum of 1 hour after the first oral dose.

At 36 hours take a CRP and Gentamicin level. At the same time give a dose of oral Amoxicillin and flush the cannula. Leave the cannula in. If the CRP is falling then remove the cannula and continue oral antibiotics. If the CRP is still rising then give the 2nd dose of gentamicin and consider repeat CRP 24 hours later. 4th dose of Benzyl penicillin should not be given as Amoxicillin has already been given.

Babies discharged on NOAH required to complete a further 5 day course of oral antibiotics (total course 7 days). This can be done at home and does not require ongoing admission in NNU or transitional care providing that they are otherwise well.

Oral antibiotic choice – Amoxicillin oral suspension 30mg/kg three times a day. For 5 days to be prescribed as a TTO in addition to the 36 hours of IV antibiotics Total antibiotic duration should be 7 days

4.2.4 Exclusion Criteria

- Any baby with positive blood, CSF or line cultures
- CRP > 50
- If a third CRP has been taken due to up trending values and shows a continued increase despite being on IV antibiotics
- Unable to obtain blood cultures from baby
- If there is a decision to treat with 2 weeks of antibiotics for meningitis
- If there any safeguarding, transport or comprehension concerns, consider remaining in hospital to complete antibiotics
- Any baby where there are ongoing clinical concerns despite IV antibiotic treatment
  - Fever or hypothermia
  - Abnormal liver or renal function on blood tests
  - Unusual rashes
  - Lethargy or poor feeding
  - Other clinical concerns

4.2.5 Advice for Parents on Discharge Home

If parents have any concern about their baby while on oral antibiotic therapy they should call the neonatal unit.

Prior to discharge parents will be given an advice leaflet. Parents may also be directed to the Amoxicillin Medicines for Children leaflet available here

Complete a discharge summary for the baby and include the EPIC smartphrase '.ORALANTIBIOTICSDISCHARGEADVICE'

Please ensure prior to discharge, that parents have been booked into the virtual follow up clinic (details below). Provide them with the date and time of their appointment before they depart from the hospital. The baby should then be added to the "Oral Antibiotic Follow-up" list on EPIC, with the date and time of their appointment written in the blue specialty note.

4.2.6 Follow Up Procedure

Any baby who has been treated with oral antibiotics at home will require a clinic follow up prior to the end of their antibiotic course. This will be through a telephone consultation with a set proforma. This will be available to view on EPIC using the Smartphrase '.ORALANTIBIOTICSWITCHREVIEW'. (Appendix 2)

This follow-up clinic will be carried out by the doctor allocated to low care. The "Oral Antibiotic Follow-up" list should be checked daily and any phone calls due should be conducted in the morning prior to completing the low care ward round.

4.3 Late-Onset Neonatal Sepsis

Late onset sepsis (LOS) is defined by onset at least 72 hours after birth with the peak incidence reported to be between the 10th and 22nd day of life. In general, the microbial characteristics and patient population are very different to those encountered in early onset sepsis and therefore this presentation requires different management.

4.3.1 Causative Organisms:

- Coagulase-negative staphylococci (CONS) – accounting for 53.2-77.9% of cases.
- Gram negative bacilli (E coli, Klebsiella, Enterobacter, Pseudomonas).
- Fungi (Candida).

4.3.2 Risk Factors for Late Onset Sepsis:

- Prematurity
- VLBW/ELBW infants
- Mechanical ventilation
- Intravascular catheterisation
- Failure of early enteral feeding with breast milk
- Prolonged duration of parenteral nutrition
- Hospitalisation
- Surgery
- Underlying respiratory and cardiovascular disease

Symptoms and signs – see 'Presenting features' Page 4

Clinical signs of LOS are generally subtle and non-specific and therefore vigilance and a high index of suspicion is required. LOS should be considered in any baby on the neonatal unit where there has been a change/deterioration in their clinical condition, which may include:

- Heart rate – reduced variability and/or decelerations
- Increasing respiratory rate and/or oxygen requirement
- Apnoeas
- Blood pressure instability
- Temperature instability
- Blood sugar instability
- Gastrointestinal symptoms: Poor feeding/Feed intolerance/ Vomiting/ Abdominal distension
- Neurological symptoms: lethargy, poor tone, bulging fontanelle, seizures (see meningitis section)

4.3.3 Investigations

Essential Initial Investigations

These investigations that should be carried out in every case of suspected LOS prior to antibiotics:

- Blood culture
- CRP – at presentation and repeated 18-24 hours later
- Full blood count
- Blood gas and glucose

Further Investigations to Consider

These investigations can be considered prior to antibiotics if there is a strong clinical suspicion of sepsis, the procedure is safe to perform in the patient and if it will not unduly delay administration of antibiotics:

- Lumbar puncture – see Neonatal Meningitis section
- Urine – suprapubic aspirate or catheter specimen
- Swabs – e.g. of umbilicus, eye, wound – as clinically indicated (see sticky eye guideline).

If the baby is clinically unwell other investigations will need to be considered depending on the symptoms present e.g. CXR, calcium, clotting studies, LFTs and bilirubin, U&Es.

4.3.4 Treatment

Initial Treatment:

If late-onset sepsis is suspected, the baby should be empirically started antibiotics which should be given within 1 hour (ideally 30 minutes) of recognition of need to treat.

Empirical antibiotics for late onset infection are IV Flucloxacillin and Gentamicin
See Neonatal Antibiotics Guideline for details including antibiotic choice for specific infection concerns

4.3.5 Fungal Prophylaxis

Give prophylactic oral Nystatin to all babies treated with antibiotics for suspected late-onset neonatal bacterial infection if they:

- have a birthweight of up to 1,500g or
- were born at less than 30 weeks' gestation.

If oral administration of nystatin is not possible, give prophylactic intravenous fluconazole

4.3.6 Ongoing Management

Ongoing management including duration of antibiotics will depend on clinical picture and microbiology advice.

- In general, antibiotics should be stopped at 36 hours if baby has:
  - Normalising clinical course,
  - Negative blood culture at 36 hours and
  - Serial negative CRPs
- If positive blood culture or specific infection identified, continue IV antibiotics as per Neonatal Antibiotic Guidelines, also taking into account:
  - Culture and sensitivity results
  - Evolving clinical picture
  - Any discussions with microbiology

4.4 Meningitis

Introduction

- Neonatal meningitis is rare it is associated with significant morbidity and mortality and therefore requires prompt recognition and appropriate management
- It has a mortality of up to 10-15% and survivors are at high risk of neurological and
- neurodevelopmental sequelae
- Neonatal meningitis may present with very subtle signs/symptoms. Those that are
- present may be indistinguishable from those seen in neonatal sepsis without meningitis

Common signs and symptoms:

- Temperature instability (present in 60%)
- Neurological symptoms – irritability (present in 60%), lethargy, poor tone, seizures (present in 20-50%), bulging fontanelle (present in 25%), nuchal rigidity (present in 15%)
- Poor feeding/vomiting (present in 50%)
- Respiratory distress (present in 33-50%)
- Apnoea (present in 10-30%)
- Diarrhoea (present in 20%)

4.4.1 Indications for Lumbar Puncture (LP):

- Clinical symptoms/signs of meningitis – consider performing LP before giving antibiotics if:
  - there is a high level of clinical suspicion of meningitis or sepsis,
  - the patient is stable ***and***
  - performing the procedure will not significantly delay antibiotics
- Blood culture positive
- Unsatisfactory response to antibiotic treatment
- A raised CRP is an indication to *consider* performing an LP:
  - **CRP <20 – LP only if clinically concerning features**
  - **CRP 20-40 – consider LP**
  - **CRP >40 – LP recommended**

4.4.2 Interpretation of LP Microscopy:

- CSF WCC > 20 cells/microlitre is suggestive of meningitis (although this diagnosis should also be considered even if the cell count is lower if the clinical presentation is suggestive)
  - Traumatic LP: CSF WCC should ***not*** routinely be adjusted for CSF RBC ratio – see references
- CSF protein and glucose levels are highly variable in neonates with and without meningitis and therefore interpretation is often difficult. In general, the following would be consistent with neonatal meningitis, but not by themselves diagnostic:
  - CSF protein >1.0g/L in term and >1.5g/L in preterm infants
  - CSF glucose <1.7mmol/L in term and <1.1mmol/L in preterm infants
- Gram stain: presence of any organism on gram stain is highly suggestive of meningitis although a negative gram stain does not exclude meningitis.

4.4.3 Indications for Repeat LP:

A repeat lumbar puncture may be considered in the presence of:

- Persistent or re-emergent fever
- Deterioration in clinical condition
- New clinical findings (especially neurological findings)
- Persistently abnormal inflammatory markers

A repeat LP is not routinely recommended prior to stopping antibiotics if the baby is making a good clinical recovery.

4.4.4 Initial Treatment:

Following a report of positive CSF microscopy (see above) antibiotics should be changed to:

- Cefotaxime *and*
- Amoxicillin

For more detail, see neonatal antibiotic guidelines

4.4.5 Corticosteroids

Use of corticosteroids is ***not*** currently recommended for cases of suspected or confirmed neonatal meningitis.

4.4.6 Follow Up for Babies with Confirmed Meningitis

- Refer directly to CHIME audiology for full audiology assessment
  - Following bacterial meningitis these babies do not have the routine hearing screening but are referred directly to audiology
  - This cannot be done via EPIC. Email referral form (appendix 3) to cse.chimereferrals@nhs.net
- Physiotherapy referral as an inpatient
- Ensure neurodevelopmental follow-up has been ordered (and confirm this by messaging the neonatal secretaries)
- Named consultant follow up at 4-6 weeks post discharge

4.5 Neonatal Antibiotic Guidelines

These antibiotic guidelines are a recommendation only and should be reviewed with the evolving clinical picture, sensitivities of reported organisms and following any discussions with the microbiology team

Indication	Antibiotic(s)	Duration
EMPIRICAL ANTIBIOTIC TREATMENT REGIMENS:		
Early Onset Neonatal Sepsis (<72hrs age) (culture pending)	Benzylpenicillin	Review with culture results
	and Gentamicin	Review with culture results
Late Onset Neonatal Sepsis (>72hrs age) (culture pending)	Flucloxacillin	Review with culture results
	and Gentamicin	Review with culture results
RESPIRATORY INFECTIONS:		
Congenital pneumonia	Benzylpenicillin	5 - 7 days
	and Gentamicin	5 - 7 days
SEPTICAEMIA WITHOUT MENINGITIS:		
GBS Septicaemia without meningitis	Benzylpenicillin	10 days
	and Gentamicin	2 days
Gram negative septicaemia without meningitis	Cefotaxime	Until sensitivities available then discuss with microbiology
	and Gentamicin	
MENINGITIS:		
Positive CSF microscopy Culture pending	Cefotaxime *and* Amoxicillin	Review with culture results
GBS or strep pneumonia meningitis	Benzylpenicillin	14 days
Listeria meningitis	Amoxicillin	21 days
	and Gentamicin	7 days
Gram negative meningitis	Cefotaxime	21 days
Positive CSF microscopy Organism unknown	Cefotaxime	14 days

SKIN/SOFT TISSUE INFECTIONS:		
Umbilical Flare / Cellulitis	Flucloxacillin	5-7 days (review with culture results and consider oral switch once clinical improvement)
ABDOMINAL SEPSIS:		
Suspected sepsis with abdominal symptoms but not definitive NEC	Flucloxacillin or Benzylpencillin (depending on age)	10-14 days
	and Gentamicin	10-14 days
	and Metronidazole	10-14 days
Necrotising enterocolitis	Amoxicillin	10 – 14 days
	and Gentamicin	10 – 14 days
	and Metronidazole	10 – 14 days

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

5.2 Relevant Policies:

- Incident Reporting Policy and Procedure.
- Claims Management Policy and Procedure.
- Policy and Procedure for the Management of Complaints, Concerns, Comments and compliments

6. REFERENCES

NICE Guideline NG195. Neonatal Infection: antibiotics for prevention and treatment. Last update March 2024 https://www.nice.org.uk/guidance/ng195

Kaiser Permanente Neonatal Sepsis Calculator https://neonatalsepsiscalculator.kaiserpermanente.org/

Joint British Association for Sexual Health and HIV and Royal College of Obstetricians and Gynaecologists national UK guideline for the management of herpes simplex virus (HSV) in pregnancy and the neonate (2024 update)

Gifford A, Wooding EL, Ng KF Is it safe for neonates with probable bacterial infection to be treated with oral antibiotics in high-income countries?*Archives of Disease in Childhood* 2024;109:681-687.

Keij FM, Kornelisse RF, Hartwig NG, *et al* RAIN study: a protocol for a randomised controlled trial evaluating efficacy, safety and cost-effectiveness of intravenous-to-oral antibiotic switch therapy in neonates with a probable bacterial infection *BMJ Open* 2019;9:e026688. doi: 10.1136/bmjopen-2018-026688

7. PUBLICATION DETAILS

Author of Clinical Guideline	Harriet Aughey, Consultant Paediatrician
Division/ Department responsible for Clinical Guideline	Women's & Children's Health, Perinatal
Contact details	01392 406631
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Replaces version number	Early onset infection Late onset infection 'Infection' guideline Meningitis Antibiotics IV to PO antibiotics Update November 2025 replaces version 1.0
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Review date	03/05/2030
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APPENDIX 1

Kaiser Neonatal Early-Onset Sepsis Calculation

PREDICTOR	SCENARIO
Incidence of Early-Onset Sepsis	0.04/1000 live births
Gestational age	weeks days
Highest maternal antepartum temperature	Celsius
ROM (Hours)	
Maternal GBS status	☐ Negative ☐ Positive ☐ Unknown
Type of intrapartum antibiotics	☐ Broad spectrum antibiotics > 4 hrs prior to birth ☐ Broad spectrum antibiotics 2-3.9 hrs prior to birth ☐ GBS specific antibiotics > 2 hrs prior to birth ☐ No antibiotics or any antibiotics < 2 hrs prior to birth

RISK PER 1000/BIRTHS
EOS Risk @ Birth

CLINICAL EXAM	
Well Appearing	☐
Equivocal	☐
Clinical Illness	☐

CLINICAL RECOMMENDATION	VITALS	
No culture, no antibiotics	Routine Vitals	☐
Blood culture	Vitals every 4 hours for 24 hours	☐
Empiric antibiotics	Vitals per NICU	☐

Signature: _____

Print Name: _____

GR-LB-NN13-01

PROOF

APPENDIX 2

Neonatal Oral Switch Telephone Review Proforma

Smartphrase on EPIC ***.ORALANTIBIOTICSWITCHREVIEW***

Telephone call with

Baby's age today @AGE@

Number of days of antibiotics so far (total including IV)

Questions for parents

How is your baby generally? Do you have any concerns?

Do you feel confident in what signs to watch out for that might indicate that your baby is becoming unwell?

{Yes/No:24880}

Is your baby showing any worrying signs? If so please describe them {Yes/No:24880}

Ask parents specifically about the following:

Unusual behaviour (e.g.- constant crying or being very quiet)
{Yes/No:24880}

Unusually floppy
{Yes/No:24880}

Develops difficulties with feeding or tolerating feeds
{Yes/No:24880}

Abnormal temperature (not explained by environmental factors) ≤36°C or ≥38°C {Yes/No:24880}

Rapid breathing or grunting {Yes/No:24880}

A change in skin colour
{Yes/No:24880}

How is your baby feeding?

Breast	{Yes/No:24880}
EBM via bottle	{Yes/No:24880}
Formula	{Yes/No:24880}
Combination	{Yes/No:24880}

How is feeding going? Any concerns?

Last weight? (please record date and weight in kg or g)

Are you having any difficulties giving your baby the antibiotic? If so please describe them (e.g. spitting it out/vomiting etc)

How many doses have you missed or have been vomited or spat out?

Do you know who to contact if you have concerns? **(NNU whilst on antibiotics, MW/GP once course complete)** {Yes/No:24880}

***** please check that this baby's details have been added to the data collection spreadsheet "P:\PAEDS SHARED\Neonatal Oral Antibiotic Switch Project 2024\Data collection for evaluation of oral switch project.xlsx"**

APPENDIX 3

Referral to Audiology

Email to cse.chimereferrals@nhs.net

Patient details -

Title/Name:	NHS No: Hospital No:
DOB:	Gender:
Address:	Home no: Mobile no: Email:
GP:	GP Address:
Name of referrer: Job Title:	Address of referrer:
Mobility issues: YES / NO If yes, please specify	Interpreter required? YES / NO If yes, please specify

Reason for referral: Date:

Reason for referral:	
	Date: